

An Improved Inference for IV Regressions^{*}

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Abstract

Empirical instrumental variables (IV) studies often report separate results based on low-dimensional instruments and many base instruments. This paper proposes a combination test that integrates these commonly reported statistics. The test linearly combines a cluster-robust Wald statistic based on low-dimensional IVs with leave-one-cluster-out Lagrangian Multiplier (LM) and Anderson-Rubin (AR) statistics constructed from many IVs. Under strong identification of the low-dimensional IVs, we establish joint asymptotic normality and asymptotic optimality of the proposed test. The procedure yields costless efficiency improvements, automatically adapts to weak identification of many instruments, and is accompanied by a practical rule of thumb for assessing efficiency gains.

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1 Introduction

Empirical applications of instrumental variables (IV) regressions in economics often involve multiple sets of candidate instruments, some having dimensionality large relative to the sample size, whereas others do not. A canonical example is the influential study by Angrist and Krueger (1991), in which they estimate the causal effect of schooling on wages using three quarter-of-birth (QoB) dummies as instruments. In pursuit of potential efficiency gain, they further interact the three QoB instruments with state- and year-of-birth dummies, yielding 180 instruments in total (the original QoB dummies plus all interactions). The paper reports estimation and inference results from both the three-instrument and the expanded 180-instrument specifications.

A different but related set of examples arises with shift-share IVs, which are now widely used in labor, public, development, macroeconomics, international trade, and finance.¹ As noted by Goldsmith-Pinkham et al. (2020), a shift-share IV can be interpreted as a particular way of combining many base IVs under appropriate conditions.² In their example following Card (2009), 38 base IVs (respective shares of immigrants from 38 home countries) are used to build the shift-share instrument for estimating the (negative) inverse elasticity of substitution between immigrants and natives in a sample of size 124. As in Angrist and Krueger (1991), the authors present results based on the one-dimensional shift-share IV as well as on the full set of base IVs. Additionally, they adopt recent empirical practice by reporting results from alternative IV estimators designed to reduce potential bias from the use of many instruments (e.g., see Table 6 in Goldsmith-Pinkham et al. (2020)).

As illustrated above, empirical studies commonly report results based on low-dimensional IVs, which typically utilize a subset or an aggregate of a potentially rich instrument set, along

¹See, for instance, Bartik (1991), Blanchard, Katz, Hall, and Eichengreen (1992), Adao, Kolesár, and Morales (2019), Goldsmith-Pinkham, Sorkin, and Swift (2020), Borusyak, Hull, and Jaravel (2022), Borusyak, Hull, and Jaravel (2025), and references therein.

²For instance, the identification strategy in Goldsmith-Pinkham et al. (2020) relies on the assumption of exogenous shares.

with results that adhere to the full set of base IVs. This naturally raises the question of whether one can systematically combine these standard estimation and inference outputs to construct improved IV inference, while retaining the respective strengths of approaches along both lines. A further practical concern is what additional steps empirical researchers must undertake to implement such a combined procedure and how much such a proposal can enhance existing methods in a way that matters empirically.

This paper aims to offer a constructive solution to the above questions on improving efficiency for IV regressions. Specifically, we consider an efficiently combined inference procedure of three commonly reported test statistics for IV regressions in an arguably general clustered setting, where the data consist of many clusters of bounded size. The core component statistics are a standard cluster-robust Wald statistic associated with the low-dimensional IVs, a leave-one-cluster-out Lagrangian Multiplier (LM) statistic, and a leave-one-cluster-out Anderson-Rubin (AR) statistic. The LM and AR statistics are based on the many base IVs, and the leave-one-cluster-out construction is employed to remove the many-IV bias in the presence of within-cluster error dependence.

Two clarifications are in order. First, we do not seek to derive the optimal test statistic by searching over all possible combinations of a given set of base IVs, nor do we combine outputs across alternative specifications that use different sets of base IVs or different low-dimensional IVs. Instead, we study how to combine existing, commonly reported test statistics derived from a given set of many IVs with their prescribed low-dimensional counterparts. Nevertheless, we attribute a notion of asymptotic optimality to the combination test in the sense of [Müller \(2011\)](#). Second, accommodating cluster sizes that grow with the sample is technically demanding in our current setting and is beyond the scope of this paper.

As it turns out, an optimal way to combine the three component test statistics is, in fact, through a linear combination of them. Specifically, we first show that the Wald, LM, and AR statistics are jointly asymptotically normal under the null hypothesis (and local alternatives), assuming that the low-dimensional IVs strongly identify the parameter of interest.

Standard optimal testing theory then implies that the uniformly most powerful unbiased (UMPU) test in the limiting problem rejects for large absolute values of an appropriate linear combination of the three limiting Gaussian observations.³ Our proposed test, as a function of the three statistics, is asymptotically that UMPU test, thus yielding a linear combination of the underlying Wald, LM, and AR statistics.

A direct consequence of the combination construction is a costless efficiency gain compared with the conventionally reported Wald test that uses only the low-dimensional IVs, since one can always place full weight on that component and disregard the information in the AR and LM statistics. Notably, the combination test adapts to the identification strength of the many base IVs. When the parameter of interest is weakly identified under many IVs in the sense of [Mikusheva and Sun \(2022\)](#), our test reduces to the component Wald test, and thus remains valid and as powerful as prior to combination.

The rationale for determining the optimal weights in the combination test is not immediately obvious. To understand this, we rely on three properties of the joint Gaussian limiting distribution of the three component statistics. First, the local alternative parameter appears in the limiting noncentralities of both the Wald and LM statistics, but with distinct scaling factors that closely reflect the relative identification strength under low-dimensional versus many IVs. Second, pairwise correlations among the three statistics are the key determinants of the covariance matrix in the Gaussian limit. Third, the scalar sufficient statistic for the local alternative parameter can be expressed as a linear combination of suitably decorrelated Gaussian components. Together with the standard UMPU testing theory, these three observations imply that the optimal weights are proportional to the relevant scaling factors and, in particular, should be less dispersed when correlations are small.

Furthermore, the confidence interval implied by the combination test has the usual “estimator plus and minus a standard error times a critical value” form. Its center is an estimator

³See, for instance, Section 4.2 in [Lehmann and Romano \(2006\)](#) and Lemma 2.2 in [Lim, Wang, and Zhang \(2024a\)](#) for the formal development of such arguments.

that linearly combines a standard GMM estimator based on the low-dimensional IVs with a leave-one-cluster-out estimator based on the many base IVs and the AR statistic, using weights that capture both the relative identification strength from the two IV sets and the UMPU weights. The efficiency gain manifests itself as an almost surely shorter confidence interval. We measure this gain by the percentage reduction in the length of the resulting confidence interval relative to that of the Wald test. As an illustration, in the immigrant enclave application of [Card \(2009\)](#), the combination procedure shortens the length by between 8% and 26%. This reduction depends mainly on the identification strengths of the low-dimensional and many IVs, together with the limiting correlations between the Wald and LM statistics and between the LM and AR statistics. In [Section 2](#), we translate these relationships into a practical rule of thumb and illustrate it using the [Card \(2009\)](#) application.

Relation to the literature. This paper contributes to a large literature on many (weak) instruments.⁴ It is especially related to [Lim et al. \(2024a\)](#), which, following [Andrews \(2016\)](#), introduces a jackknife conditional linear combination (CLC) test that is robust to weak identification, many instruments, and heteroskedasticity. They propose a linear combination of jackknife AR, jackknife LM, and orthogonalized jackknife LM tests to ensure good power performance under different identification scenarios. Their analysis, however, focuses on many-IV settings, and hence is not directly applicable to our framework. In contrast, we also incorporate low-dimensional IVs, which are commonly encountered in practice, in a more

⁴See, for instance, [Kunitomo \(1980\)](#), [Morimune \(1983\)](#), [Bekker \(1994\)](#), [Donald and Newey \(2001\)](#), [Chao and Swanson \(2005\)](#), [Stock and Yogo \(2005\)](#), [Han and Phillips \(2006\)](#), [Andrews and Stock \(2007\)](#), [Hansen, Hausman, and Newey \(2008\)](#), [Newey and Windmeijer \(2009\)](#), [Anderson, Kunitomo, and Matsushita \(2010\)](#), [Kuersteiner and Okui \(2010\)](#), [Anatolyev and Gospodinov \(2011\)](#), [Okui \(2011\)](#), [Belloni, Chen, Chernozhukov, and Hansen \(2012\)](#), [Carrasco \(2012\)](#), [Chao, Swanson, Hausman, Newey, and Woutersen \(2012\)](#), [Hausman, Newey, Woutersen, Chao, and Swanson \(2012\)](#), [Kolesár \(2013\)](#), [Hansen and Kozbur \(2014\)](#), [Carrasco and Tchuente \(2015\)](#), [Wang and Kaffo \(2016\)](#), [Kolesár \(2018\)](#), [Evdokimov and Kolesár \(2018\)](#), [Sølvsten \(2020\)](#), [Chao, Swanson, and Woutersen \(2023\)](#), [Lim et al. \(2024a\)](#), [Boot and Nibbering \(2024\)](#), [Yap \(2024\)](#), among others.

general clustered setting. Under our assumption of strong identification for low-dimensional IVs, the resulting test remains adaptive to the strength of many base IVs. Similarly, in the context of causal inference under covariate-adaptive randomization, [Jiang, Li, Miao, and Zhang \(2025\)](#) proposes a new estimator of average treatment effect (ATE) using an optimal linear combination of estimators with and without regression adjustments.

In settings with many instruments and clustered data, how to perform robust estimation and inference is not straightforward. [Frandsen, Leslie, and McIntyre \(2025\)](#) propose a cluster-robust jackknife IV estimator (CJIVE) and show that it remains consistent under many instruments and clustering, but they do not derive a consistent cluster-robust variance estimator or a valid inference procedure. As pointed out by [Chao et al. \(2012\)](#), when the number of (possibly weak) instruments is large, both linear and quadratic components contribute to the asymptotic variance, and the quadratic component may dominate. The standard cluster-robust two-stage least squares (TSLS) variance estimator based on a jackknifed instrument,⁵ as implemented in Stata and commonly used in practice, neglects the quadratic term and therefore becomes invalid when many instruments are present.

On the other hand, [Ligtenberg \(2023\)](#) develops cluster jackknife AR and LM tests that are robust to weak identification, many instruments, and clustering. In the same spirit, our paper invokes cluster-robust variance estimators for AR and LM statistics in IV models with many instruments, a fixed number of controls, and clustered data. Recent work further indicates that suitably constructed bootstrap methods can produce more reliable inference than those based on asymptotic approximations in IV models under heteroskedasticity or clustering.⁶

⁵This constructed one-dimensional IV predicts unit i 's endogenous variable by leaving out i 's own observation or i 's entire cluster; see, e.g., Section 3.2 of [Chyn, Frandsen, and Leslie \(2024\)](#) in the case of judge design. The resulting IV estimator coincides numerically with JIVE or CJIVE, so both linear and quadratic variance terms must be taken into account.

⁶See, for instance, [Davidson and MacKinnon \(2010\)](#), [Finlay and Magnusson \(2019\)](#), [Lim, Wang, and Zhang \(2024b\)](#), [MacKinnon \(2021\)](#), and [Wang and Zhang \(2024\)](#), among others.

Lastly, as highlighted by [Chao et al. \(2023\)](#), conducting estimation and inference is challenging when both the number of instruments and the number of controls are large.⁷ In this paper, we focus on models with a fixed number of controls and leave the case of many-IVs-many-controls for future research.

Structure of the paper: Section 2 details the rationale behind the proposed rule of thumb and demonstrates its use in an empirical application; practitioners mainly interested in applications may focus on this section directly. Section 3 introduces the model and key preliminaries, while Section 4 develops the large-sample theory for the combination test and formalizes its theoretical properties. Section 5 provides additional simulation results, and Section 6 concludes. An additional case with weakly identified low-dimensional IVs alongside strongly identified many IVs, as well as all proofs, is presented in the Online Appendices.

Notation. We write $[n] \equiv \{1, \dots, n\}$ and $[G] \equiv \{1, \dots, G\}$. Let A be an $n \times m$ matrix and let $\{n_g\}_{g \in [G]}$ be positive integers with $\sum_{g=1}^G n_g = n$. We denote by $A_{[g]}$ the g -th row-wise block of A , of dimension $n_g \times m$. When A is a $n \times n$ square matrix, we denote by $A_{[g,h]}$ the (g, h) -th block of A . For a positive semi-definite square matrix A , denote its largest and smallest eigenvalues by $\lambda_{\max}(A)$ and $\lambda_{\min}(A)$, respectively. Let C be a generic positive constant independent of n , whose value may change from line to line. For brevity, we write $\sum_{g,h \in [G]^2, g \neq h} := \sum_{g \in [G]} \sum_{h \in [G], h \neq g}$.

2 Rule of Thumb and Empirical Illustration

In this section, we develop a practical rule of thumb that can be directly applied to reported estimates and standard errors from regressions using low-dimensional IVs, as well as from regressions employing many IVs. We illustrate its empirical relevance using the empirical

⁷For advances in estimation and inference methods in such high-dimensional IV and control settings, see [Akerberg and Devereux \(2009\)](#), [Kolesár \(2013\)](#), [Evdokimov and Kolesár \(2018\)](#), [Chao et al. \(2023\)](#), [Mikusheva and Sun \(2024\)](#), [Boot and Nibbering \(2024\)](#), and [Yap \(2024\)](#), among others.

application in [Goldsmith-Pinkham et al. \(2020, Section VII\)](#), which builds on [Card \(2009\)](#).

As outlined in the Introduction, we measure the efficiency gain from the combination test by the percentage reduction in the length of its confidence interval relative to that of the Wald test in large samples. [Section 4.3](#) will formally establish two key properties of this measure. First, there is no reduction in interval length only when the relative identification strength of the many IVs to the low-dimensional IVs exactly matches the limiting correlation between the corresponding Wald and LM statistics. Consequently, the combination test almost always yields a shorter confidence interval, so we recommend that practitioners routinely perform the additional step of combination inference.

However, the percentage reduction generally depends on several primitive quantities: the identification strengths of both low-dimensional and many IVs, as well as the limiting correlations between the Wald and LM statistics and between the LM and AR statistics. This complexity makes it difficult to convert the reduction directly into a simple practical rule. The second property addresses this issue: the percentage reduction is monotonically increasing in the absolute correlation between the LM and AR statistics, which implies a lower bound on the efficiency gain that depends only on the correlation between the Wald and LM statistics and the ratio of the standard deviations of the standard GMM estimator using low-dimensional IVs to the leave-one-cluster-out estimator using the many IVs (or, equivalently, the relative strength of the many IVs to the low-dimensional IVs).

[Figure 1](#) plots the lower bound as a function of the standard deviation ratio for different values of ρ_1 , the limiting correlation between the Wald and LM statistics. Two observations emerge. First, for any fixed ρ_1 , we show theoretically that whenever the standard deviation ratio exceeds ρ_1 , the lower bound on efficiency gains increases with the ratio, reflecting the fact that the combination test exploits the additional precision provided by the LM statistic. Second, once the standard deviation ratio exceeds one, the lower bound decreases as ρ_1 increases, because the LM statistic then contributes relatively little additional information beyond the highly correlated Wald statistic. As a simple rule of thumb that only requires a

back of envelope calculation based on the reported standard errors, we propose: for empirically plausible values of ρ (between -0.7 and 0.7), whenever the standard error from the regression with low-dimensional IVs divided by that from the regression with many IVs is greater than 1.05 , the corresponding confidence interval is reduced by at least 10% .⁸

To illustrate the empirical relevance of efficiency gains and the associated rule of thumb, we implement the combination test in an empirical application that estimates the (negative) inverse elasticity of substitution between immigrants and natives, following Card (2009). As in Goldsmith-Pinkham et al. (2020, Section VII), we examine two separate sets of results by skill group: high school equivalent workers and college equivalent workers. The analysis is based on cross-sectional regressions for each skill group in the year 2000 in 124 cities. The dependent variable is the residual log wage gap between immigrant and native men, and the main regressor of interest is the log ratio of immigrant to native hours of both men and women within the same skill group. Because a positive labor demand shock to immigrants can simultaneously increase their earnings and labor supply relative to natives, this can introduce a potential endogeneity. However, such a shock draws immigrants into a location disproportionately relative to natives, thus motivating the use of a Bartik instrument to address endogeneity. To construct the Bartik instrument, immigration shares from 38 countries (groups) in 1980 are used as the base instruments, and the final instrument is formed as a weighted average of these country-specific shares, where the weights are given by the number of arrivals to the United States between 1990 and 2000 by origin group and skill group (see Goldsmith-Pinkham et al. (2020) for further details).

Table 1 reports the point estimates obtained from the standard two-stage least squares (TSLS) estimator using the Bartik instrument, $\hat{\beta}_1$, along with those from the leave-one-out estimator, $\hat{\beta}_2$, which relies on all base IVs. We present results separately for specifications

⁸In additional (unreported) plots for $\rho \in [-0.99, 0.99]$, the confidence interval is at least 10% shorter whenever the standard deviation ratio exceeds 1.11 (with corresponding thresholds 1.05 for 5% and 1.25 for 20%).

that include and exclude city-level controls. As expected and consistent with the findings in Goldsmith-Pinkham et al. (2020), the estimates are broadly similar within each skill group. However, in every specification, the confidence intervals constructed from the TSLS estimator (Wald CI) differ to some extent from those based on $\hat{\beta}_2$ (LM CI). This discrepancy is partly due to the different standard errors of the two estimators, which we exploit in our combination test to obtain strictly shorter confidence intervals across all specifications. For example, for workers with college equivalent skills, our confidence interval is roughly 8% and 26% shorter, respectively, than the Wald CI in the specifications with and without controls.

Figure 2 displays the realized percentage reduction in confidence interval length achieved by our combination test, together with the theoretical lower bounds for that reduction, analogous to Figure 1, but calculated using the specification-specific estimate $\hat{\rho}_1$ of the limiting correlation between the Wald and LM statistics. Two observations are worth noting. First, the theoretical lower bounds closely track the actual percentage reductions across specifications, even though these bounds are derived solely from the Wald and LM statistics. It indicates that, in this empirical setting, combining the AR statistic yields little additional efficiency gain. This aligns with our earlier theoretical discussion that efficiency gains increase monotonically with the absolute correlation between LM and AR statistics, and, in fact, the corresponding consistent estimates $\hat{\rho}_2$ are not particularly large in any specification.

Second, Figure 2 clearly illustrates our proposed rule of thumb. All estimated $\hat{\rho}_1$ values fall within $[-0.7, 0.7]$. When the standard error ratio exceeds 1.05, the actual reduction in the length of the confidence interval is 26% (specification without controls for college equivalent workers), much above the rule-of-thumb benchmark 10%. In contrast, when the standard error ratio remains at or below 1.05, the improvements are modest, reflecting the converse of our rule of thumb. Nevertheless, even in these cases, our combination test can still deliver confidence intervals that are about 8% (specification with controls for college equivalent workers), 5% (specification without controls for high school equivalent workers) and 1% (specification with controls for high school equivalent workers) shorter.

3 Model and Preliminaries

3.1 Setup

We consider a clustered dataset with G clusters and denote the size of the g -th cluster as n_g for $g \in [G]$. We index observations by clusters followed by units. Denote $I_g = [N_{g-1} + 1, \dots, N_g]$, where $N_0 = 0$, $N_g = \sum_{g'=0}^g n_{g'}$, and $N_G = n$. Then, $\{I_g\}_{g \in [G]}$ forms a partition of $[n]$, and if $i \in I_g$, this means that the i -th observation belongs to the g -th cluster. We then consider a linear IV regression with clustered data

$$\tilde{Y}_{i,g} = \tilde{X}_{i,g}\beta + W_{i,g}^\top \gamma + \tilde{e}_{i,g}, \quad (3.1)$$

where we denote $\tilde{Y}_{i,g} \in \mathbb{R}$, $\tilde{X}_{i,g} \in \mathbb{R}$, and $W_{i,g} \in \mathbb{R}^{d_w}$ as an outcome variable, an endogenous regressor, and exogenous regressors, respectively. Further denote $\tilde{Z}_{i,g} \in \mathbb{R}^K$ as the IVs for $\tilde{X}_{i,g}$. The first-stage equation can be written as

$$\tilde{X}_{i,g} = \tilde{\Pi}_{i,g} + \tilde{V}_{i,g}, \quad (3.2)$$

where $\tilde{\Pi}_{i,g} = \mathbb{E}(X_{i,g} | \{\tilde{Z}_{j,g}, W_{j,g}\}_{j \in I_g})$ is not assumed to be linear in $\tilde{Z}_{i,g}$ and $W_{i,g}$. We assume that $\mathbb{E}\tilde{e}_{i,g} = 0$ and $\mathbb{E}\tilde{V}_{i,g} = 0$, and $\{\tilde{e}_{i,g}, \tilde{V}_{i,g}\}_{i \in I_g, g \in [G]}$ are independent between clusters, but allow them to have a general dependence structure within each cluster. Throughout the paper, the dimension d_w of $W_{i,g}$ is assumed to be fixed. If researchers want to include cluster fixed effects in the model, they can obtain (3.1) by first demeaning the data (outcome, endogenous regressor, controls, and instruments) at the cluster level. We assume that K , the dimension of $\tilde{Z}_{i,g}$, diverges to infinity with the sample size, while d_w remains fixed.

Let $\tilde{Y}$, $\tilde{X}$, $\tilde{\Pi}$, W , $\tilde{Z}$ be $n \times 1$, $n \times 1$, $n \times 1$, $n \times d_w$, and $n \times K$ -dimensional vectors and matrices formed by $\tilde{Y}_{i,g}$, $\tilde{X}_{i,g}$, $\tilde{\Pi}_{i,g}$, $W_{i,g}$, and $\tilde{Z}_{i,g}$, respectively. More specifically, $\tilde{Y}$ is constructed by stacking up $\tilde{Y}_{i,g}$ across $i \in I_g$ followed by $g \in [G]$, and similarly for $\tilde{X}$, $\tilde{\Pi}$, W and $\tilde{Z}$. We then partial out W from $\tilde{Y}$, $\tilde{X}$, and $\tilde{Z}$, so that the model in (3.1)-(3.2) can be

written in a vector form as

$$Y = X\beta + e, \quad X = \Pi + V, \quad (3.3)$$

where $Y = M_W \tilde{Y}$, $X = M_W \tilde{X}$, $\Pi = M_W \tilde{\Pi}$, $e = M_W \tilde{e}$, $V = M_W \tilde{V}$, $M_W = I_n - P_W$, $P_W = W(W^\top W)^{-1}W^\top$ and I_n denotes an $n \times n$ identity matrix. We further denote $Z = M_W \tilde{Z}$.

In addition, besides the K -dimensional many IVs $\tilde{Z}_{i,g} \in \mathbb{R}^K$, we assume that there is another set of low-dimensional IVs

$$\tilde{z}_{i,g} = f_{i,g}(\tilde{Z}, W) \in \mathbb{R}^{d_z},$$

where $\{f_{i,g}(\cdot)\}_{i \in I_g, g \in [G]}$ is a list of known nonstochastic functions of d_z dimension. Specifically, as illustrated by the example of [Angrist and Krueger \(1991\)](#) in the Introduction, researchers may begin with certain low-dimensional base IVs $\tilde{z}_{i,g}$, such as the three QoB dummies, and construct a large number of new IVs by taking the interaction between the base IVs and control variables $W_{i,g}$ (e.g., state- and year-of-birth dummies in [Angrist and Krueger \(1991\)](#)). Then, $\tilde{z}_{i,g}$ is a subset of the K -dimensional many IVs $Z_{i,g}$ for the model in (3.3), which include both the low-dimensional base IVs and interacted IVs. The second example of $\tilde{z}_{i,g}$ is the widely used shift-share IV. As pointed out by [Goldsmith-Pinkham et al. \(2020\)](#), under their identification strategy that treats the shares as exogenous, the one-dimensional shift-share IV can be regarded as a weighted average of high-dimensional base IVs, i.e., $\tilde{z}_{i,g} = \sum_{k=1}^K \tilde{Z}_{i,g} h_{i,k}$. For instance, in the study of China shock by [Autor, Dorn, and Hanson \(2013\)](#), the observations are clustered at the level of US commuting zone with a short panel data setting. Then, according to our notation, the employment share of the US industry k in the commuting zone g in a certain initial time period $i = 0$ corresponds to the base IVs $\tilde{Z}_{0,g}$ (i.e., $\tilde{Z}_{i,g} = Z_{0,g}$ for all i), the supply shock from China to the US industry k in the time period i corresponds to the weight $h_{i,k}$, and K is the total number of industries. Let $\tilde{z}$ be the $n \times d_z$ -dimensional matrix formed by $\tilde{z}_{i,g}$, and denote $z = M_W \tilde{z}$. In many

empirical applications, the dimension d_z is just one, but our setup also allows for $d_z > 1$, while maintaining the requirement that d_z is fixed with respect to the sample size n .

We focus on the model with a scalar endogenous variable for two reasons. First, in many empirical applications of IV regressions, there is only one endogenous variable (as can be seen from the surveys by [Andrews, Stock, and Sun \(2019\)](#) and [Lee, McCrary, Moreira, and Porter \(2022\)](#)). Second, as we assume at least the low-dimensional IVs provide strong identification, our results can be extended to testing of scalar restrictions with multiple endogenous variables by applying standard subvector inference methods, without appealing to a projection-based inference approach.⁹ If the identification strength for many IVs is mixed, our method could potentially be extended by following the approach of [Chao et al. \(2012\)](#). We leave this extension for future research.

The null and alternative hypotheses studied are $\mathcal{H}_0 : \beta = \beta_0$ against $\mathcal{H}_1 : \beta \neq \beta_0$.

3.2 Test Statistics

It is possible to conduct inference directly based on the low-dimensional IVs. Specifically, given a $d_z \times d_z$ positive definite weighting matrix $\hat{A}_n$, the generalized method of moments (GMM) estimator can be written as

$$\hat{\beta}_1 = (X^\top z \hat{A}_n z^\top X)^{-1} (X^\top z \hat{A}_n z^\top Y).$$

⁹For weak-identification-robust subvector inference, in general, one may use a projection approach ([Dufour and Taamouti, 2005](#)) after implementing inference on the whole vector of endogenous variables. However, the projection approach typically leads to conservative inference. Alternative subvector inference methods for IV regressions (e.g., see [Guggenberger, Kleibergen, Mavroeidis, and Chen \(2012\)](#), [Andrews \(2017\)](#), [Guggenberger, Kleibergen, and Mavroeidis \(2019, 2021\)](#), and [Wang and Doko Tchatoka \(2018\)](#)) provide a power improvement over the projection approach under a fixed number of instruments (some of these methods further require conditional homoskedasticity). However, whether they can be applied to the setting of many weak instruments is unclear.

It is also possible to construct test statistics using the K -dimensional IVs (i.e., the many IVs). Denote $P = Z(Z^\top Z)^{-1}Z^\top$ as the projection matrix of Z . Then, the leave-one-cluster-out estimator of β is denoted as $\hat{\beta}_2$ and defined as

$$\begin{aligned}\hat{\beta}_2 &= \left(\sum_{g,h \in [G]^2, g \neq h} X_{[g]}^\top P_{[g,h]} X_{[h]} \right)^{-1} \left(\sum_{g,h \in [G]^2, g \neq h} X_{[g]}^\top P_{[g,h]} Y_{[h]} \right) \\ &= (X^\top (P - \bar{P})X)^{-1} (X^\top (P - \bar{P})Y),\end{aligned}$$

where $\bar{P}$ is the block diagonal matrix corresponding to P such that the g -th block on its diagonal is $P_{[g,g]}$.

Given $\hat{\beta}_1$ and $\hat{\beta}_2$, we define the estimator $\hat{\beta}$ as

$$\hat{\beta} = \frac{\ddot{\Phi}_2^{1/2}}{\dot{\Phi}_1^{1/2} + \ddot{\Phi}_2^{1/2}} \times \hat{\beta}_1 + \frac{\dot{\Phi}_1^{1/2}}{\dot{\Phi}_1^{1/2} + \ddot{\Phi}_2^{1/2}} \times \hat{\beta}_2,$$

where $\dot{\Phi}_1$ and $\ddot{\Phi}_2$ are the variance estimators for $\hat{\beta}_1$ and $\hat{\beta}_2$, respectively, to be defined later.

We show that $\hat{\beta}$ is consistent whenever either the low-dimensional IV estimator $\hat{\beta}_1$ or the many-IV estimator $\hat{\beta}_2$ is consistent. Because researchers do not need to know which of the two estimators is consistent when constructing $\hat{\beta}$, the estimator $\hat{\beta}$ is doubly robust.

We then use the doubly robust estimator $\hat{\beta}$ to re-estimate the variances associated with $\hat{\beta}_1$ and $X^\top (P - \bar{P})(Y - X\beta)$, denoted by $\hat{\Phi}_1$ and $\hat{\Sigma}$, respectively, and defined later. These variance estimates are used to construct the Wald statistic and the leave-one-cluster-out jackknife Lagrange multiplier (LM) statistic,

$$\begin{aligned}T(\beta_0) &= \frac{(X^\top z \hat{A}_n z^\top X)^{-1} X^\top z \hat{A}_n z^\top e(\beta_0)}{\sqrt{\hat{\Phi}_1}} = \frac{\hat{\beta}_1 - \beta_0}{\sqrt{\hat{\Phi}_1}}, \\ LM(\beta_0) &= \frac{X^\top (P - \bar{P})e(\beta_0)}{\sqrt{\hat{\Sigma}}},\end{aligned}$$

where $e(\beta_0) = Y - X\beta_0$.

Lastly, as pointed out by [Hausman et al. \(2012\)](#), [Lim et al. \(2024a\)](#), and [Mikusheva and Sun \(2024\)](#), it is possible to use the jackknife AR statistic to further improve the efficiency of the jackknife LM statistic. In the case with clustered data, we define the leave-one-cluster-out jackknife AR statistic as

$$AR = \frac{\hat{e}^\top (P - \bar{P}) \hat{e}}{\sqrt{\hat{\Upsilon}}}, \quad (3.4)$$

where $\hat{\Upsilon}$ is a consistent variance estimator for the numerator defined later, and $\hat{e} = Y - X\hat{\beta}$.

We use the consistent estimator $\hat{\beta}$ instead of the null hypothesis β_0 to construct the AR statistic for at least two reasons: (1) the two choices are asymptotically equivalent under strong identification and local alternatives; and (2) as shown by [Lim et al. \(2024a\)](#), even under strong identification and fixed alternatives, the optimal combination test based on the jackknife LM and AR statistics—when the AR is constructed using β_0 —can lead to a non-monotonic power curve, while this can be avoided if the AR statistic is constructed using $\hat{\beta}$. Consequently, the AR statistic in (3.4) does not depend on the null hypothesis β_0 . Instead, it should be viewed as a normalized estimator of zero, which is used solely to improve the efficiency of our procedure.

In the next section, we illustrate how to optimally combine the three test statistics $T(\beta_0)$, $LM(\beta_0)$, and AR . The corresponding variance estimators $(\dot{\Phi}_1, \ddot{\Phi}_2, \hat{\Phi}_1, \hat{\Sigma}, \hat{\Upsilon})$ are introduced in [Section 4.1](#).

3.3 Combination Test

Given the three test statistics $(T(\beta_0), LM(\beta_0), AR)$, we seek to combine them in a theoretically justified way that can improve on the Wald test based only on the low-dimensional IVs. The key insight of our paper is that under certain local alternative $\beta - \beta_0 = \delta d_n$ with

some deterministic sequence $d_n \downarrow 0$, we have the following joint limiting distribution:

$$\begin{pmatrix} T(\beta_0) \\ LM(\beta_0) \\ AR \end{pmatrix} \rightsquigarrow \mathcal{N} \left(\begin{pmatrix} a_1 \delta \\ a_2 \delta \\ 0 \end{pmatrix}, \begin{pmatrix} 1 & \rho_1 & 0 \\ \rho_1 & 1 & \rho_2 \\ 0 & \rho_2 & 1 \end{pmatrix} \right) \quad (3.5)$$

for some a_1 , a_2 , ρ_1 and ρ_2 to be defined later. In this limiting problem, the UMPU level- α test for the default null hypothesis $\delta = 0$ against two-sided alternatives, which are solely based on the limiting three-dimensional normal random vector, can be obtained by invoking standard hypothesis-testing results (see, for example, Section 4.2 of [Lehmann and Romano \(2006\)](#)), and is stated in Proposition 3.1 below.

Proposition 3.1. *Suppose one observes $(\mathcal{N}_1, \mathcal{N}_2, \mathcal{N}_3)$, which follows the limiting distribution in (3.5) with $\rho_1^2 + \rho_2^2 < 1$ and wants to test $\mathcal{H}_0 : \delta = 0$ against $\mathcal{H}_1 : \delta \neq 0$ for known values of $(a_1, a_2, \rho_1, \rho_2)$, then UMPU level- α test rejects if*

$$\left(\frac{b_1 \tilde{\mathcal{N}}_1 + b_2 \tilde{\mathcal{N}}_2 + b_3 \tilde{\mathcal{N}}_3}{\sqrt{b_1^2 + b_2^2 + b_3^2}} \right)^2 \geq \mathbb{C}_\alpha,$$

where $\mathbb{C}_\alpha$ is the $(1-\alpha)$ percentile of a chi-squared random variable with one degree of freedom,

$$\begin{pmatrix} \tilde{\mathcal{N}}_1 \\ \tilde{\mathcal{N}}_2 \\ \tilde{\mathcal{N}}_3 \end{pmatrix} = \begin{pmatrix} 1 & \rho_1 & 0 \\ \rho_1 & 1 & \rho_2 \\ 0 & \rho_2 & 1 \end{pmatrix}^{-1/2} \begin{pmatrix} \mathcal{N}_1 \\ \mathcal{N}_2 \\ \mathcal{N}_3 \end{pmatrix}, \text{ and } \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix} = \begin{pmatrix} 1 & \rho_1 & 0 \\ \rho_1 & 1 & \rho_2 \\ 0 & \rho_2 & 1 \end{pmatrix}^{-1/2} \begin{pmatrix} a_1 \\ a_2 \\ 0 \end{pmatrix}.$$

The corresponding power function for the UMPU test is

$$\mathbb{P} \left(\chi_1^2 \left(\delta^2 \frac{(1 - \rho_2^2)a_1^2 - 2\rho_1 a_1 a_2 + a_2^2}{1 - \rho_1^2 - \rho_2^2} \right) \geq \mathbb{C}_\alpha \right),$$

where $\chi_1^2(\lambda)$ is a noncentral chi-squared with noncentrality λ and one degree of freedom.

In light of this optimal testing result in the limiting problem, one may wish to propose

implementing the following test:

$$\phi_n^o = \mathbf{1} \{ (\omega_1 T(\beta_0) + \omega_2 LM(\beta_0) + \omega_3 AR)^2 \geq \mathbb{C}_\alpha \} \equiv \phi^* (T(\beta_0), LM(\beta_0), AR), \quad (3.6)$$

$$\text{where } \begin{pmatrix} \omega_1 \\ \omega_2 \\ \omega_3 \end{pmatrix} = \frac{1}{\sqrt{b_1^2 + b_2^2 + b_3^2}} \begin{pmatrix} 1 & \rho_1 & 0 \\ \rho_1 & 1 & \rho_2 \\ 0 & \rho_2 & 1 \end{pmatrix}^{-1/2} \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix}, \quad (3.7)$$

and then investigate its asymptotic justification.

However, the parameters a_1, a_2, ρ_1, ρ_2 are usually unknown and need to be estimated. In addition, it turns out that the weights $(\omega_1, \omega_2, \omega_3)$ are invariant to the scale normalization of (b_1, b_2, b_3) , and thus, (a_1, a_2) . Therefore, to construct the UMPU test, it suffices to consistently estimate $\alpha_1 = a_1/\sqrt{a_1^2 + a_2^2}$ and $\alpha_2 = a_2/\sqrt{a_1^2 + a_2^2}$ along with ρ_1 and ρ_2 .

Given the consistent estimators $(\hat{\alpha}_1, \hat{\alpha}_2, \hat{\rho}_1, \hat{\rho}_2)$ for $(\alpha_1, \alpha_2, \rho_1, \rho_2)$ specified in Section 4.1, we then implement the feasible version of the combination test:

$$\phi_n^* = \mathbf{1} \{ (\hat{\omega}_1 T(\beta_0) + \hat{\omega}_2 LM(\beta_0) + \hat{\omega}_3 AR)^2 \geq \mathbb{C}_\alpha \}, \quad (3.8)$$

$$\text{where } \begin{pmatrix} \hat{\omega}_1 \\ \hat{\omega}_2 \\ \hat{\omega}_3 \end{pmatrix} = \frac{1}{\sqrt{\hat{b}_1^2 + \hat{b}_2^2 + \hat{b}_3^2}} \times \begin{pmatrix} 1 & \hat{\rho}_1 & 0 \\ \hat{\rho}_1 & 1 & \hat{\rho}_2 \\ 0 & \hat{\rho}_2 & 1 \end{pmatrix}^{-1/2} \begin{pmatrix} \hat{b}_1 \\ \hat{b}_2 \\ \hat{b}_3 \end{pmatrix}, \quad \begin{pmatrix} \hat{b}_1 \\ \hat{b}_2 \\ \hat{b}_3 \end{pmatrix} = \begin{pmatrix} 1 & \hat{\rho}_1 & 0 \\ \hat{\rho}_1 & 1 & \hat{\rho}_2 \\ 0 & \hat{\rho}_2 & 1 \end{pmatrix}^{-1/2} \begin{pmatrix} \hat{\alpha}_1 \\ \hat{\alpha}_2 \\ 0 \end{pmatrix}.$$

4 Large-Sample Theory

In this section, we investigate the asymptotic behavior of our combination test. We begin by stating and discussing general assumptions about the data-generating process and on the identification strength of both the low-dimensional and many IVs. We then establish the asymptotic efficiency properties of the combination test and, finally, compare its efficiency to that of the conventional Wald test based solely on low-dimensional IVs via the limiting

length ratio of their confidence intervals.

4.1 General Assumptions

As in [Chao et al. \(2012\)](#), we treat $\tilde{Z}$ and W as fixed. This is equivalent to treating them as random and repeating all the analyses in the paper by conditioning on them. For the data-generating process, we impose the following assumptions.

Assumption 1. *The following conditions hold when n is sufficiently large:*

1. $\max_{i \in I_g, g \in [G]} \mathbb{E}(\tilde{e}_{i,g}^4 + \tilde{V}_{i,g}^4) \leq C < \infty;$

2. $\max_{1 \leq g \leq G} n_g \leq C < \infty;$

3. Let

$$\Omega_g = \mathbb{E} \left[\begin{pmatrix} \tilde{e}_{[g]} \tilde{e}_{[g]}^\top & \tilde{e}_{[g]} \tilde{V}_{[g]}^\top \\ \tilde{V}_{[g]} \tilde{e}_{[g]}^\top & \tilde{V}_{[g]} \tilde{V}_{[g]}^\top \end{pmatrix} \right], \quad 1 \leq g \leq G,$$

then

$$0 < \frac{1}{C} \leq \min_{1 \leq g \leq G} \lambda_{\min}(\Omega_g) \leq \max_{1 \leq g \leq G} \lambda_{\max}(\Omega_g) \leq C < \infty;$$

- 4.

$$0 < \frac{1}{C} \leq \lambda_{\min} \left(\frac{1}{n} \sum_{i \in I_g, g \in [G]} z_{i,g} z_{i,g}^\top \right) \leq \lambda_{\max} \left(\frac{1}{n} \sum_{i \in I_g, g \in [G]} z_{i,g} z_{i,g}^\top \right) \leq C < \infty,$$

$$0 < \frac{1}{C} \leq \lambda_{\min} \left(\frac{1}{n} \sum_{i \in I_g, g \in [G]} W_{i,g} W_{i,g}^\top \right) \leq \lambda_{\max} \left(\frac{1}{n} \sum_{i \in I_g, g \in [G]} W_{i,g} W_{i,g}^\top \right) \leq C < \infty,$$

and

$$\max_{i \in I_g, g \in [G]} \left\{ \left| \tilde{\Pi}_{i,g} \right| + |\Pi_{i,g}| \right\} \leq C < \infty, \quad \max_{i \in I_g, g \in [G]} \left\{ \|z_{i,g}\|_2 + \|W_{i,g}\|_2 \right\} = o(\sqrt{n}).$$

Remark 4.1. Assumption 1.1 is a standard condition on the moments of error terms. Assumption 1.2 restricts the cluster size to be bounded, which incorporates cross-sectional and short-panel data structures. It is also possible to extend our analysis to the case with divergent cluster sizes, especially when the within-cluster dependence is weak. However, when the cluster size is allowed to diverge, and there is strong within-cluster dependence, the convergence rates of various IV estimators depend on the identification strength, the within-cluster dependence, the cluster sizes, and the number of clusters in a complicated way. We leave the investigation in this direction for future research. Assumption 1.3 ensures that the error covariance matrix is non-singular for each cluster. Finally, Assumption 1.4 is a mild condition for the design matrix.

For the low-dimensional IVs, we focus on the case of strong identification strength. We investigate the case in which the low-dimensional IVs have weak identification strength in Supplementary Appendix A, and show that, under certain conditions, when the many-IV has strong identification, our optimal combination test still asymptotically controls size. The cluster-robust variance estimator $\hat{\Phi}_1$ for the Wald statistic is defined as

$$\begin{aligned}\hat{\Phi}_1 &= (X^\top z \hat{A}_n z^\top X)^{-1} (X^\top z \hat{A}_n \hat{\Omega} \hat{A}_n z^\top X) (X^\top z \hat{A}_n z^\top X)^{-1}, \quad \text{and} \\ \hat{\Omega} &= \sum_{g \in [G]} (z_{[g]}^\top \hat{e}_{[g]}) (z_{[g]}^\top \hat{e}_{[g]})^\top.\end{aligned}$$

The initial estimator $\dot{\Phi}_1$ for Φ_1 used in the computation of $\hat{\beta}$ is defined in the same way as $\hat{\Phi}_1$, except that $\hat{e}_{[g]} = Y_{[g]} - X_{[g]}\hat{\beta}$ is replaced by $\dot{e}_{[g]} = Y_{[g]} - X_{[g]}\hat{\beta}_1$.

We make the following assumptions regarding the inference with low-dimensional IVs.

Assumption 2. *The following conditions hold almost surely:*

1. *There exists a sequence of non-random positive definite matrices A_n such that*

$$A_n^{-1/2} \hat{A}_n A_n^{-1/2} \xrightarrow{p} I_{d_z},$$

and $\lambda_{\max}(A_n)/\lambda_{\min}(A_n) \leq C < \infty$ for all n large enough;

2. Let $r_n = \|z^\top \Pi\|_2$, then $\sqrt{n}/r_n \rightarrow 0$.

Remark 4.2. Assumption 2.1 states that the weighing matrix $\hat{A}_n$ converges in probability to some non-random positive definite matrix, which is standard in the GMM setup. Assumption 2.2 ensures that $z^\top \Pi$, the deterministic part of $z^\top X$, dominates $z^\top V$, the random part of $z^\top X$, and is therefore the key condition for the consistency of $\hat{\beta}_1$ for β (i.e., the low-dimensional IVs provide strong identification).

We do not require the identification strength provided by the many IVs to be strong, in the sense that $\Pi^\top \Pi / \sqrt{K} \rightarrow \infty$ (e.g., see Mikusheva and Sun (2022)). In Section 4.2 below, we show that the optimal combination test is adaptive to the identification strength provided by the many IVs. In particular, it controls size asymptotically and has non-trivial power even when $\hat{\beta}_2$ is inconsistent.

If the many-IV identification is strong, similar to Chao et al. (2012), we can show that the asymptotic variance of $\hat{\beta}_2$ is

$$\Phi_2 = (\Pi^\top (P - \bar{P}) \Pi)^{-1} \Sigma (\Pi^\top (P - \bar{P}) \Pi)^{-1},$$

where

$$\Sigma = \mathbb{E} \left(\sum_{g, h \in [G]^2, g \neq h} \Pi_{[g]}^\top P_{[g, h]} \left(\sum_{k \in [G]} M_{W, [h, k]} \tilde{e}_{[k]} \right) \right)^2 + \mathbb{E} \left(\sum_{g, h \in [G]^2, g \neq h} \tilde{V}_{[g]}^\top P_{[g, h]} \tilde{e}_{[h]} \right)^2$$

is the asymptotic variance of $X^\top (P - \bar{P}) e$. A natural estimator for Φ_2 is

$$\hat{\Phi}_2 = (X^\top (P - \bar{P}) X)^{-1} \hat{\Sigma} (X^\top (P - \bar{P}) X)^{-1},$$

where $\hat{\Sigma}$ is a consistent estimator of Σ . Such an estimator is proposed in Chao et al. (2012), but here we also need to account for the fact that W has already been partialled

out, whereas in [Chao et al. \(2012\)](#) the coefficients for W are also estimated. Therefore, some adjustments are required, as in [Matsushita and Otsu \(2024\)](#). For that purpose, define $Q = M_W(P - \bar{P})M_W$, and let $\bar{Q}$ be the block diagonal matrix corresponding to Q such that the g -th block on its diagonal is $Q_{[g,g]}$. Our variance estimator is similar to the one in [Chao et al. \(2012\)](#) but with $P - \bar{P}$ replaced by $Q - \bar{Q}$, i.e.,

$$\hat{\Sigma} = \sum_{g \in [G]} \left(\sum_{h \in [G], h \neq g} \tilde{X}_{[h]}^\top Q_{[h,g]} \hat{e}_{[g]} \right)^2 + \sum_{g, h \in [G]^2, g \neq h} \left(\tilde{X}_{[g]}^\top Q_{[g,h]} \hat{e}_{[h]} \right) \left(\tilde{X}_{[h]}^\top Q_{[h,g]} \hat{e}_{[g]} \right).$$

The initial estimator $\ddot{\Phi}_2$ for Φ_2 used in the computation of $\hat{\beta}$ is defined in the same way as $\hat{\Phi}_2$, except that $\hat{e}_{[g]} = Y_{[g]} - X_{[g]}\hat{\beta}$ is replaced by $\ddot{e}_{[g]} = Y_{[g]} - X_{[g]}\hat{\beta}_2$.

Last, for the jackknife AR statistic, the variance estimator is given by

$$\hat{\Upsilon} = 2 \sum_{g, h \in [G]^2, g \neq h} \left(\hat{e}_{[g]}^\top P_{[g,h]} \hat{e}_{[h]} \right)^2,$$

which is consistent for the asymptotic variance of $\hat{e}^\top (P - \bar{P}) \hat{e}$, given by

$$\Upsilon = \mathbb{E} \left(\sum_{g, h \in [G]^2, g \neq h} \tilde{e}_{[g]}^\top P_{[g,h]} \tilde{e}_{[h]} \right)^2.$$

We make the following assumptions regarding the inference with many IVs.

Assumption 3. *The following conditions hold:*

1. $K \rightarrow \infty$ as $n \rightarrow \infty$ such that $\limsup_{n \rightarrow \infty} K/n \leq C < 1$;
2. $\text{rank}(P) = K$ and $\max_{1 \leq g \leq G} \lambda_{\max}(P_{[g,g]}) \leq C < 1$;
3. Let $\hat{\Pi} = M_W(P - \bar{P})\Pi = Q\tilde{\Pi}$ and $\bar{\Pi} = (Q - \bar{Q})\tilde{\Pi}$, then

$$\max_{i \in I_g, g \in [G]} \left\{ |\hat{\Pi}_{i,g}| + |\bar{\Pi}_{i,g}| \right\} \leq C < \infty,$$

and

$$\tilde{\Pi}^\top \tilde{\Pi} \leq C \Pi^\top \Pi, \quad \hat{\Pi}^\top \hat{\Pi} \geq \Pi^\top \Pi / C, \quad \Pi^\top (P - \bar{P}) \Pi \geq \Pi^\top \Pi / C,$$

when n is large enough;

4. For all sufficiently large n ,

$$\left| \text{corr} \left(\sum_{g,h \in [G]^2, g \neq h} \tilde{V}_{[g]}^\top P_{[g,h]} \tilde{V}_{[h]}, \sum_{g,h \in [G]^2, g \neq h} \tilde{V}_{[g]}^\top P_{[g,h]} \tilde{e}_{[h]} \right) \right| \leq C < 1.$$

Remark 4.3. Assumption 3.1 allows the dimension of many IVs K to be proportional to the sample size n . Assumption 3.2 is similar to the standard condition that $\max_{1 \leq i \leq n} P_{ii} \leq C < 1$ in the literature on many instruments, and the restriction that $\text{rank}(P) = K$ will exclude redundant columns from Z . Assumption 3.3 holds in general if $\tilde{\Pi}^\top \tilde{\Pi}$, $\Pi^\top \Pi$ and $\Pi' P \Pi$ are of the same order. Assumption 3.4 excludes perfect correlations between the two quadratic forms. Note that we do not put any restriction on the identification strength of many IVs, i.e., we allow $\Pi^\top \Pi / \sqrt{K}$ to be bounded.

4.2 Asymptotic Efficiency Properties of Combination Test

We now investigate the asymptotical properties of ϕ_n^* when the low-dimensional IVs are strong, in the sense that Assumption 2 holds. This allows us to define the local alternative according to the asymptotic variance of $\hat{\beta}_1$ and $\hat{\beta}_2$ and the limiting covariance structure of the component statistics, from which the joint limiting distribution of $(T(\beta_0), LM(\beta_0), AR)$ can be derived. The results with weak low-dimensional IVs are given in Supplementary Appendix A. The formal regularity condition is stated as follows.

Assumption 4. *The following limits exist*

$$\begin{aligned}\rho_1 &= \lim_{n \rightarrow \infty} \frac{1}{\sqrt{\Psi\Sigma}} \sum_{g \in [G]} \mathbb{E} \left[\left(\dot{\Pi}_{[g]}^\top \tilde{e}_{[g]} \right) \left(\hat{\Pi}_{[g]}^\top \tilde{e}_{[g]} \right) \right], \\ \rho_2 &= \lim_{n \rightarrow \infty} \frac{2}{\sqrt{\Sigma\Upsilon}} \sum_{g, h \in [G]^2, g \neq h} \mathbb{E} \left[\left(\tilde{V}_{[g]}^\top P_{[g, h]} \tilde{e}_{[h]} \right) \left(\tilde{e}_{[g]}^\top P_{[g, h]} \tilde{e}_{[h]} \right) \right],\end{aligned}$$

where $\dot{\Pi} = zA_n z^\top X$ and $\rho_1^2 + \rho_2^2 < 1$.

Remark 4.4. The asymptotic expansion of $LM(\beta_0)$ includes both linear and quadratic functions of the errors $(\tilde{e}, \tilde{V})$, whereas AR depends only on a quadratic function of $\tilde{e}$. The linear and quadratic components are asymptotically normal and uncorrelated, and therefore asymptotically independent. Since the Wald statistic $T(\beta_0)$ involves only linear functions of the errors, it is asymptotically uncorrelated with AR . Finally, ρ_1 and ρ_2 denote, respectively, the correlation between the linear components of $T(\beta_0)$ and $LM(\beta_0)$, and the correlation between the quadratic components of $LM(\beta_0)$ and AR .

The following theorem establishes the joint distribution of the three test statistics above under the local alternative.

Theorem 4.1. *Under Assumptions 1–4, suppose that there exists a deterministic sequence $d_n \downarrow 0$ such that $d_n \Phi_1^{-1/2} \rightarrow a_1$, $d_n \Phi_2^{-1/2} \rightarrow a_2$, and $\beta - \beta_0 = \delta d_n$ for some fixed δ , where $a_1 \geq 0$, $a_2 \geq 0$ and $a_1^2 + a_2^2 > 0$. Then we have the joint limiting distribution (3.5) for $(T(\beta_0), LM(\beta_0), AR)'$.*

Remark 4.5. The existence of the sequence d_n is ensured by Assumption 2. In particular, we may define $d_n = \min(\Phi_1^{1/2}, \Phi_2^{1/2})$. Under the strong identification of the low-dimensional IVs in Assumption 2.2, we have $\Phi_1^{1/2} = O\left(\frac{\sqrt{n}}{r_n}\right) = o(1)$, which immediately implies $d_n = \min(\Phi_1^{1/2}, \Phi_2^{1/2}) = o(1)$, regardless of the order of Φ_2 .

Remark 4.6. The joint normality established in Theorem 4.1 holds even when the many-IV specification is weakly identified in the sense of Mikusheva and Sun (2022), that is, when

$\Pi^\top \Pi / \sqrt{K}$ is bounded. This result follows from two observations. First, the estimator $\hat{\beta}$ used to construct the variance estimators $\hat{\Phi}_2$ and $\hat{\Upsilon}$ for the LM and AR statistics remains consistent due to the strong identification of the low-dimensional IVs and the double robustness of $\hat{\beta}$. Second, under weak many-IV identification, the quadratic components of the LM and AR statistics dominate their asymptotic behavior and yield asymptotic normality as long as $K \rightarrow \infty$. In this regime, we have $\rho_1 = 0$, $d_n = \sqrt{n}/r_n$, and $\Phi_2^{-1} = O(1)$, which further imply that $a_2 = 0$ under Assumption 2.2.

To implement the optimal test ϕ_n^* defined in (3.8), we still need to estimate a_1 , a_2 , ρ_1 , and ρ_2 . It turns out to be easier to estimate $\alpha_1 = a_1 / \sqrt{a_1^2 + a_2^2}$ and $\alpha_2 = a_2 / \sqrt{a_1^2 + a_2^2}$, and we propose the following estimators

$$\hat{\alpha}_1 = \frac{\sqrt{\hat{\Phi}_2}}{\sqrt{\hat{\Phi}_1 + \hat{\Phi}_2}}, \quad \hat{\alpha}_2 = \frac{\sqrt{\hat{\Phi}_1}}{\sqrt{\hat{\Phi}_1 + \hat{\Phi}_2}}.$$

In addition, let $\hat{X} = M_W(P - \bar{P})X$ and $\hat{X} = z\hat{A}_n z^\top X$, for ρ_1 and ρ_2 we propose the following estimators

$$\begin{aligned} \hat{\rho}_1 &= \frac{1}{\sqrt{\hat{\Psi}\hat{\Sigma}}} \sum_{g \in [G]} \left[\left(\hat{X}_{[g]}^\top \hat{e}_{[g]} \right) \left(\hat{X}_{[g]}^\top \hat{e}_{[g]} \right) \right], \\ \hat{\rho}_2 &= \frac{2}{\sqrt{\hat{\Sigma}\hat{\Upsilon}}} \sum_{g, h \in [G]^2, g \neq h} \left[\left(\hat{X}_{[g]}^\top P_{[g, h]} \hat{e}_{[h]} \right) \left(\hat{e}_{[g]}^\top P_{[g, h]} \hat{e}_{[h]} \right) \right]. \end{aligned}$$

By combining Proposition 3.1 and Theorem 4.1, and invoking the approach developed by Müller (2011), we obtain a precise sense of asymptotic optimality for our proposed test ϕ_n^* , which is formalized in the following Theorem 4.2.

Theorem 4.2. *Let $\mathcal{M}$ denote the set of data generating processes m that satisfy the conditions of Theorem 4.1 pointwise for all $\delta \in \mathfrak{R}$. Suppose that one wants to test $\mathcal{H}_0 : \delta = 0$*

against $\mathcal{H}_1 : \delta \neq 0$. Then, for the class $\mathfrak{C}$ of tests ϕ_n satisfying that

$$\lim_{n \rightarrow \infty} \mathbb{E}[\phi_n] \leq \alpha \quad \text{for all } m \in \mathcal{M}, \delta = 0, \quad (4.1)$$

$$\liminf_{n \rightarrow \infty} \mathbb{E}[\phi_n] \geq \alpha \quad \text{for all } m \in \mathcal{M}, \delta \neq 0, \quad (4.2)$$

we have $\phi_n^* \in \mathfrak{C}$, and, for any $\delta_1 \neq 0$ and any $\phi_n \in \mathfrak{C}$,

$$\lim_{n \rightarrow \infty} \mathbb{E}[\phi_n] \leq \lim_{n \rightarrow \infty} \mathbb{E}[\phi_n^*] \quad \text{for all } m \in \mathcal{M}, \delta = \delta_1. \quad (4.3)$$

Moreover, for the test $\tilde{\phi}_n = \mathbf{1}\{T^2(\beta_0) \geq \mathbb{C}_\alpha\}$, we have $\tilde{\phi}_n \in \mathfrak{C}$, and for any δ and all $m \in \mathcal{M}$,

$$\lim_{n \rightarrow \infty} \mathbb{E}[\tilde{\phi}_n] = \lim_{n \rightarrow \infty} \mathbb{E}[\phi_n^*] \quad \text{if and only if} \quad a_2 = \rho_1 a_1.$$

Remark 4.7. Theorem 4.2 shows that, under local alternatives, ϕ_n^* attains the asymptotic efficiency bound within the class of tests that remain asymptotically unbiased and valid for all data generating processes inducing the same weak limit for $(T(\beta_0), LM(\beta_0), AR)'$ as in Theorem 4.1. This class of tests includes, in particular, the Wald and LM tests based solely on the low-dimensional and many-IVs statistics, respectively. It also includes the HLIM and HFUL tests of Hausman et al. (2012), both of which are asymptotically equivalent to linear combinations of the LM and AR tests.

However, our optimal test ϕ_n^* does not dominate tests that cannot be expressed directly as functions of $(T(\beta_0), LM(\beta_0), AR)'$, such as the sup-score test of Belloni et al. (2012) and the ridge-regularized AR test of Dovì, Kock, and Mavroeidis (2024). Indeed, it is possible to construct data generating processes under which either the optimal combination test, the sup-score test, or the ridge-regularized AR test achieves the highest power. We establish this notion of optimality primarily to provide guidance for constructing tests that improve upon the conventional Wald test, rather than to identify a globally optimal procedure. That said,

one could potentially combine our ϕ_n^* test with the sup-score test, in the spirit of Navjeevan (2024), to obtain more powerful inference.

Remark 4.8. Theorem 4.2 also clarifies the necessary and sufficient condition under which the combination test does not deliver a strict power gain over the Wald test, namely $a_2 = \rho_1 a_1$. Recall that a_1 and a_2 represent the orders of the concentration parameters for the low-dimensional and many IVs, respectively. As will be shown below, even when we allow either of them to be zero—thereby covering situations where one IV estimator dominates the other in terms of convergence rate—the condition ($a_2 = \rho_1 a_1$) is still seldom met. Put differently, one should generally anticipate strictly powerful inference when using our combination test.

In particular, when $a_1 = 0$ and $a_2 > 0$, corresponding to the case when the identification strength of many IVs is larger than that of the low-dimensional IVs, we obtain a strict power improvement for all values of ρ_1 and ρ_2 satisfying $\rho_1^2 + \rho_2^2 < 1$. Conversely, when $a_1 > 0$ and $a_2 = 0$, meaning that low-dimensional IVs provide stronger identification than many IVs, we still achieve a strict power gain as long as $\rho_1 \neq 0$. When $a_1 > 0$ and $a_2 > 0$, that is, when the two sets of IVs have identification strengths of the same order, strict power improvement is ensured, provided that $\rho_1 \neq a_2/a_1$. Indeed, at $\rho_1 = a_2/a_1$, the sufficient statistic for δ derived from the joint limiting distribution of the three component statistics (cf. (3.5)) becomes independent of the limiting Gaussian observations associated with the LM and AR statistics, so it is not surprising that combining the Wald statistic with them does not yield a more powerful inference.

Remark 4.9. As long as the low-dimensional IVs are strongly identified, the optimal combination test ϕ_n^* does not lose asymptotic power for any degree of identification strength of the many IVs. In this sense, the efficiency gains delivered by the combination test are essentially a “free lunch.” Under local alternatives, the weak convergence result in Theorem 4.1 holds uniformly, regardless of whether the many IVs are strong or weak. In particular, when the many IVs are weak so that $a_2 = \rho_1 = 0$ (as the quadratic term in $LM(\beta_0)$ dominates the linear term), the second part of Theorem 4.2 shows that the combination test asymptotically

reduces to the Wald test, implying no efficiency loss from combining. Moreover, as shown below, the combination test remains consistent irrespective of the identification strength of the many IVs.

Interestingly, in the alternative setting where the low-dimensional IVs are weakly identified but the many IVs are strongly identified—so that the Wald test becomes invalid—our combination test continues to be asymptotically valid, provided certain suitably adapted, yet still mild conditions hold. We relegate the technical discussions of these scenarios to Supplementary Appendix A.

Finally, for any fixed alternative, both $T(\beta_0)$ and $LM(\beta_0)$ are consistent and, by construction, avoid the issue of non-monotonic power when their corresponding set of IVs is strong. Hence, it is reasonable to anticipate that our combined test will retain these desirable properties, a result that we formalize in the theorem below. However, we emphasize once more that these results remain valid regardless of the strength of the many IVs.

Theorem 4.3. *Suppose that Assumptions 1-4 hold. Then, under $\beta - \beta_0 = \delta$ for some fixed $\delta \neq 0$, we have $\lim_{n \rightarrow \infty} \mathbb{E}[\phi_n^*] = 1$.*

4.3 Quantifying Efficiency Improvement in Large Samples

We measure the efficiency improvement of the combination test over the conventional Wald test that uses only low-dimensional IVs by the percentage reduction in the asymptotic length of the resulting confidence interval. Recall from Remark 4.9 that, under weak many instruments, the combination test is asymptotically equivalent to the Wald test. Consequently, the associated confidence interval takes the usual “estimator plus and minus a standard error times a critical value” form: $[\hat{\beta}_1 - \sqrt{\hat{\Phi}_1} \times \sqrt{\mathcal{C}_\alpha}, \hat{\beta}_1 + \sqrt{\hat{\Phi}_1} \times \sqrt{\mathcal{C}_\alpha}]$, where $\hat{\beta}_1$ and $\hat{\Phi}_1$ are as defined above, and $\sqrt{\mathcal{C}_\alpha}$ is the standard normal critical value. In this case, the asymptotic efficiency gain is zero.

When the many IVs provide strong identification, the confidence interval associated with

our combination test can also be expressed in the familiar “estimator plus and minus a standard error times a critical value” form. To see this, observe that under strong identification for many IVs, the component LM statistic can be represented as follows:

$$LM(\beta_0) = \frac{X^\top (P - \bar{P})e(\beta_0)}{\sqrt{\hat{\Sigma}}} = \frac{\hat{\beta}_2 - \beta_0}{\sqrt{\hat{\Phi}_2}} + o_p(1),$$

where the $o_p(1)$ term comes from the fact that under strong identification,

$$\text{sign}(X^\top (P - \bar{P})X) \xrightarrow{p} 1.$$

Inserting this into (3.8) gives the resulting form of our combination test

$$\phi_n^* = \mathbf{1} \left\{ \left(\hat{\omega}_1 \frac{\hat{\beta}_1 - \beta_0}{\sqrt{\hat{\Phi}_1}} + \hat{\omega}_2 \frac{\hat{\beta}_2 - \beta_0}{\sqrt{\hat{\Phi}_2}} + \hat{\omega}_2 o_p(1) + \hat{\omega}_3 AR \right)^2 \geq \mathbb{C}_\alpha \right\}.$$

The resulting confidence interval is asymptotically equivalent to

$$CI^* = \left[\hat{\beta}^* - \frac{1}{\left(\hat{\omega}_1/\sqrt{\hat{\Phi}_1} + \hat{\omega}_2/\sqrt{\hat{\Phi}_2} \right)} \sqrt{\mathbb{C}_\alpha}, \hat{\beta}^* + \frac{1}{\left(\hat{\omega}_1/\sqrt{\hat{\Phi}_1} + \hat{\omega}_2/\sqrt{\hat{\Phi}_2} \right)} \sqrt{\mathbb{C}_\alpha} \right], \quad (4.4)$$

where $\hat{\beta}^*$ is a combined estimator of β ,

$$\hat{\beta}^* = \frac{\hat{\omega}_1/\sqrt{\hat{\Phi}_1}}{\left(\hat{\omega}_1/\sqrt{\hat{\Phi}_1} + \hat{\omega}_2/\sqrt{\hat{\Phi}_2} \right)} \hat{\beta}_1 + \frac{\hat{\omega}_2/\sqrt{\hat{\Phi}_2}}{\left(\hat{\omega}_1/\sqrt{\hat{\Phi}_1} + \hat{\omega}_2/\sqrt{\hat{\Phi}_2} \right)} \hat{\beta}_2 + \frac{\hat{\omega}_3 AR}{\left(\hat{\omega}_1/\sqrt{\hat{\Phi}_1} + \hat{\omega}_2/\sqrt{\hat{\Phi}_2} \right)}.$$

The intuition behind the combined estimator is fundamentally efficiency-driven. First, $\hat{\Phi}_1$ and $\hat{\Phi}_2$ estimate the asymptotic variances of $\hat{\beta}_1$ and $\hat{\beta}_2$, respectively, so for given weights $(\hat{\omega}_1, \hat{\omega}_2)$ the combined estimator $\hat{\beta}^*$ assigns greater weight to the estimator with the faster convergence rate, or equivalently, the smaller asymptotic variance. Second, although the AR

statistic is asymptotically centered at zero and therefore does not affect the location of the combined estimator, its correlation with the many-IV estimator $\hat{\beta}_2$ allows it to reduce the combined estimator's variance. This mechanism parallels the construction of the HLIML and HFUL estimators in Hausman et al. (2012). Third, the estimated weights $(\hat{\omega}_1, \hat{\omega}_2, \hat{\omega}_3)$ are consistent for the population weights $(\omega_1, \omega_2, \omega_3)$ in (3.7) associated with the UMPU test. Finally, the optimal weights defined in (3.7) solve the following optimization problem:

$$\min_{\omega_1, \omega_2, \omega_3} \frac{1}{(a_1\omega_1 + a_2\omega_2)^2} \quad \text{s.t.} \quad (\omega_1, \omega_2, \omega_3) \begin{pmatrix} 1 & \rho_1 & 0 \\ \rho_1 & 1 & \rho_2 \\ 0 & \rho_2 & 1 \end{pmatrix} \begin{pmatrix} \omega_1 \\ \omega_2 \\ \omega_3 \end{pmatrix} = 1.$$

The quadratic constraint ensures that CI^* attains the correct asymptotic coverage, while the objective corresponds to the asymptotic variance of the combined estimator $\hat{\beta}^*$, since

$$\frac{1}{d_n^2 (\omega_1/\sqrt{\Phi_1} + \omega_2/\sqrt{\Phi_2})^2} \rightarrow \frac{1}{(a_1\omega_1 + a_2\omega_2)^2}.$$

Thus, the optimal weights in (3.7), originally motivated by the UMPU testing problem, also yield an optimally efficient combined estimator of $(\hat{\beta}_1, \hat{\beta}_2, AR)$ that attains the minimal asymptotic variance.

A direct implication of the confidence interval in (4.4) is that, asymptotically, the percentage reduction in its length relative to the confidence interval based on the conventional Wald test can be derived analytically as follows,

$$\begin{aligned} & \lim_{n \rightarrow \infty} 1 - \frac{1/\left(\hat{\omega}_1/\sqrt{\hat{\Phi}_1} + \hat{\omega}_2/\sqrt{\hat{\Phi}_2}\right)}{\sqrt{\hat{\Phi}_1}} \\ &= 1 - \sqrt{\frac{(1 - \rho_1^2 - \rho_2^2)a_1^2}{(1 - \rho_2^2)a_1^2 - 2\rho_1 a_1 a_2 + a_2^2}} \end{aligned} \tag{4.5}$$

$$\geq 1 - \sqrt{\frac{(1 - \rho_1^2)a_1^2}{a_1^2 - 2\rho_1 a_1 a_2 + a_2^2}} = 1 - \sqrt{\frac{(1 - \rho_1^2)}{(1 - \rho_1^2) + (\rho_1 - a_2/a_1)^2}}. \tag{4.6}$$

As equation (4.5) shows, the efficiency gain is primarily determined by the identification strength of the low-dimensional and many IVs, as well as by the limiting correlations between the component Wald and LM statistics and between the LM and AR statistics. In particular, consistent with Theorem 4.2, when $a_2 = \rho_1 a_1$, the combination test ϕ_n^* does not yield efficiency improvement, and its confidence interval is asymptotically of the same length as that based on $\tilde{\phi}_n$. This also turns out to encompass the weak many-IV case, where $\rho_1 = a_2 = 0$. On the other hand, whenever $a_2 \neq \rho_1 a_1$, the resulting confidence interval is strictly shorter, implying improved efficiency. Moreover, the efficiency gain measure (4.5) increases monotonically in the magnitude of the correlation parameter ρ_2 , which implies the lower bound in (4.6). Figure 1 depicts this bound as a function of a_2/a_1 —the ratio of the standard deviations of $\hat{\beta}_1$ and $\hat{\beta}_2$ —for various values of ρ_1 . This leads to the practical rule of thumb, as discussed in Section 2: for empirically plausible values of ρ_1 (between -0.7 and 0.7), whenever the ratio of already reported standard errors $\sqrt{\hat{\Phi}_1}$ and $\sqrt{\hat{\Phi}_2}$ exceeds 1.05 , the associated confidence interval shrinks by at least 10% .

5 Simulation Study

In this section, we perform simulations to evaluate the finite-sample behavior of our combination test. In particular, we consider the following model with clustered data,

$$\begin{aligned}\bar{Y}_{i,g} &= \bar{X}_{i,g}\beta + \bar{W}_{i,g}^\top\gamma + \alpha_g + \bar{e}_{i,g}, \\ \bar{X}_{i,g} &= \bar{Z}_{i,g}^\top\pi + \bar{W}_{i,g}^\top\tau + \xi_g + \bar{V}_{i,g},\end{aligned}$$

where α_g and ξ_g are cluster-level fixed effects. These fixed effects are generated by $\alpha_g = u_{1g} + g/G$ and $\xi_g = u_{2g} + g/G$, $g = 1, \dots, G$, where u_{1g} and u_{2g} are independent standard normal random variables. By demeaning at the cluster level, we partial out the fixed effects and obtain $\tilde{Y}_{i,g}$, $\tilde{X}_{i,g}$, $\tilde{W}_{i,g}$, $\tilde{Z}_{i,g}$, $\tilde{V}_{i,g}$, and $\tilde{e}_{i,g}$, following the notation in Section 3.1. The control variables in $\tilde{W}_{i,g}$ are generated by the standard normal distribution, and the dimension of

$\bar{W}$ is fixed at $d_w = 10$. The instruments in $\bar{Z}_{i,g}$ are normally distributed with mean 0 and cluster-level dependence: within each cluster g , the covariance matrix is given by

$$\Omega_{1g} = \begin{bmatrix} 1 & \theta_1 & \cdots & \theta_1 \\ \theta_1 & 1 & \cdots & \theta_1 \\ \vdots & \vdots & \ddots & \vdots \\ \theta_1 & \theta_1 & \cdots & 1 \end{bmatrix}_{n_g \times n_g}, \quad g = 1, \dots, G,$$

and between clusters these instruments are independent of each other; we set $\theta_1 = 0.5$ in our simulations. Finally, to obtain an arguably complex error structure, we first generate $\acute{e}_{i,g} = \rho \varepsilon_{i,g} + \sqrt{1 - \rho^2} \sigma_{i,g} v_g$ and $\acute{V}_{i,g} = \rho \eta_{i,g} + \sqrt{1 - \rho^2} \sigma_{i,g} v_g$, where $\sigma_{i,g} = \sqrt{(0.2 + (\bar{W}_{i,g}^\top \tau)^2) / 2.4}$. Here, $\varepsilon_{i,g}$, $\eta_{i,g}$, and v_g are mutually independent standard normal random variables, ρ governs the degree of endogeneity, τ is specified below, and we fix $\rho = 0.5$ in all simulations. Next, within each cluster, we premultiply the vectors $\acute{e}_{i,g}$ and $\acute{V}_{i,g}$ by

$$\Omega_{2g} = \begin{bmatrix} 1 & 0 & \cdots & 0 \\ \theta_2 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ \theta_2^{n_i-1} & \theta_2^{n_i-2} & \cdots & 1 \end{bmatrix}_{n_g \times n_g}, \quad g = 1, \dots, G,$$

thereby generating $\bar{e}_{i,g}$ and $\bar{V}_{i,g}$. In our simulations, we set $\theta_2 = 0.7$.

For the parameters, we set $\beta = 0.3$, $\gamma = \tau = (1/\sqrt{d_w}) \times \iota_{d_w}$, where ι_{d_w} is a $d_w \times 1$ vector of ones. We specify geometrically decaying coefficients of IVs as $\pi = (\phi^0, \phi^1, \dots, \phi^{K-1})$, where K denotes the number of (many) base IVs and ϕ controls the relative weight assigned to each instrument. It is noted that $\phi = 0$ represents the case in which only the first instrument has identification strength, while $\phi = 1$ corresponds to the case in which each instrument has the same identification strength. We further normalize π to have $\|\pi\|_2 = \sqrt{\psi \sqrt{K}/n}$, where ψ controls the identification strength of the many IVs. The one-dimensional IV

is constructed by taking the average of the many IVs; as ϕ approaches one, the identification strength of the low-dimensional IV becomes stronger since it is closer to the optimal instrument. We set the sample size at $n = 2,000$ and the number of clusters at $G = 500$, and then generate heterogeneous cluster sizes following a procedure similar to that in [Djogbenou, MacKinnon, and Nielsen \(2019\)](#). Specifically, for $g = 1, \dots, G - 1$, we set $n_g = \max \left\{ 1, n \exp(2g/G) / (1 + \sum_{g=1}^{G-1} \exp(2g/G)) \right\}$ and then the size of the last cluster as $n_G = \max \left\{ 1, n - \sum_{g=1}^{G-1} n_g \right\}$. For the dimension of $\bar{Z}$, we consider $K = 100$ and $K = 500$, respectively. All the results below are based on 5,000 simulations.

Figure 3 displays the power curves for our combination test ϕ_n^* along with those for the component Wald and jackknife LM tests, at different values of K (the dimension of the many IVs), ψ (which governs the identification strength of the many IVs), and ϕ (which controls the identification strength of the one-dimensional IV relative to the many IVs). We identified three main observations, each of which aligns with our large-sample theory. First, in every scenario, the combination test ϕ_n^* attains the correct size and is more powerful than each of the other two tests. In particular, as shown in Panel C, the power curve ϕ_n^* is never dominated by that of the Wald test, regardless of the strength of the many IVs, thereby underscoring the “free lunch” efficiency gains delivered by our combination test. Second, within Panels A and B of Figure 3, we observe that for fixed K and ψ , the power improvement of ϕ_n^* over the Wald test becomes more substantial as the identification strength of the one-dimensional IV weakens relative to that of the many IVs (i.e. as ϕ decreases). This is reflected in the widening gap between the power curves of ϕ_n^* and the Wald test. It emphasizes how many IVs-based LM and AR statistics contribute to the power enhancement. Third, in contrast to the cases in which the one-dimensional IV dominates many IVs in strength (first figure in Panel A or B) and the power curve of ϕ_n^* coincides with that of the Wald test, noticeable gaps persist between the power curves of ϕ_n^* and the LM test in the flipped cases (second and third figures in Panel A or B). These gaps highlight how the AR component contributes to power enhancement through its correlation with the LM statistic.

6 Conclusion

This paper introduces a simple inference approach to improve conventionally reported estimation and inference results in instrumental variables regressions, based on low-dimensional instruments (such as aggregated shift-share IVs) and their underlying full set of (many) base instruments. The proposed combination inference procedure synthesizes three core statistics: the cluster-robust Wald statistic derived from low-dimensional IVs, and the leave-one-cluster-out Lagrangian Multiplier (LM) and Anderson-Rubin (AR) statistics derived from many base IVs.

Under strong identification of the low-dimensional instruments, we show that the component statistics are jointly asymptotically normal. The resulting combination test is constructed as an optimal linear combination of the three core statistics, based on an optimal testing result in the associated limiting experiment. In this way, the test achieves a form of asymptotic optimality in the sense of [Müller \(2011\)](#). A defining feature of this procedure is its adaptability; the test automatically weights the component statistics according to their relative identification strengths and dependence structure. Consequently, the procedure yields a “free lunch” in terms of efficiency: it is never asymptotically less powerful than standard Wald inference and, in most cases, produces strictly shorter confidence intervals. Crucially, the test adapts to the strength of the many instruments and reduces to conventional Wald inference when the many IVs are only weakly identified.

The simulation results, together with an empirical illustration based on [Card \(2009\)](#), confirm that the resulting efficiency gains can be quantitatively and empirically relevant in practice. Because the procedure is straightforward to implement and requires little beyond the standard reported outputs, we recommend the routine use of the combination test in empirical IV applications. We leave to future research the extension of our results to settings with many control variables, the consideration of alternative bootstrap procedures, and potential combinations with, for example, the sup-score test to further enhance power.

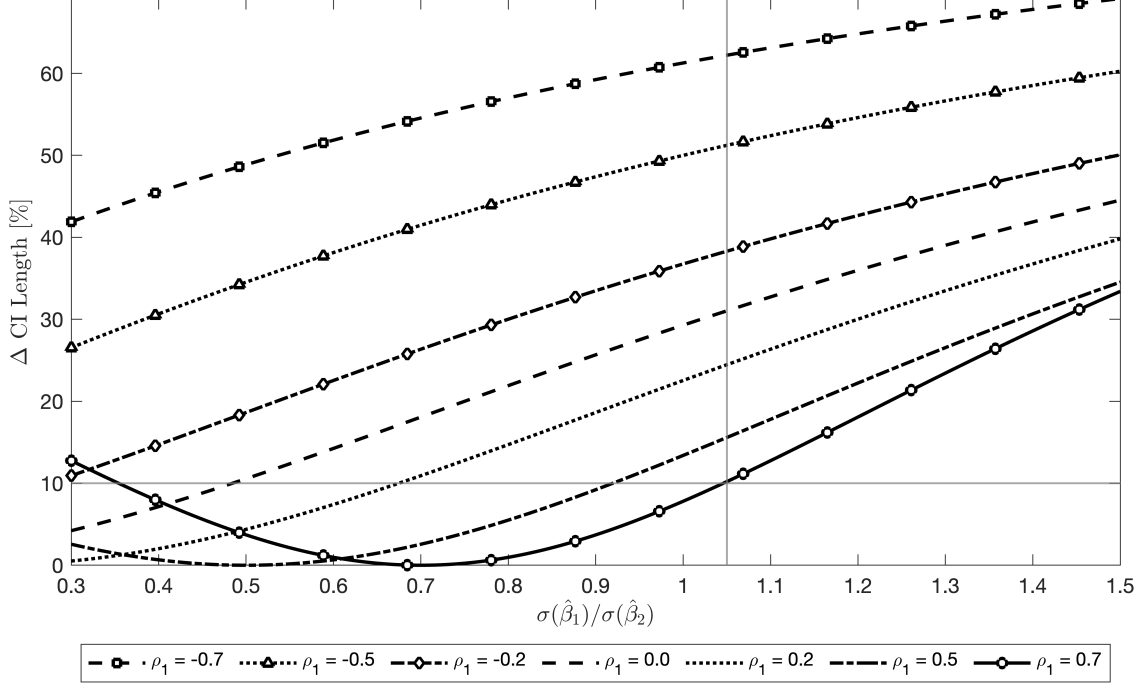


Figure 1: Theoretical lower bounds for percentage reduction in confidence interval length.

Notes: This figure plots the lower bound of the efficiency gain, given by (4.6), as a function of the standard deviation ratio, $\sigma(\hat{\beta}_1)/\sigma(\hat{\beta}_2)$, for various values of ρ_1 , the limiting correlation between the Wald and LM statistics. The horizontal axis is the ratio of standard deviation of $\hat{\beta}_1$, the standard GMM estimator using low-dimensional IVs, and standard deviation of $\hat{\beta}_2$, the leave-one-cluster-out estimator using the many base IVs. The vertical axis is the reduction in the length of confidence interval in percentage points.

	College Equivalent Workers		High school Equivalent Workers	
	Yes	No	Yes	No
$\hat{\rho}_1$	0.588	0.446	0.408	0.576
$\hat{\rho}_2$	0.103	0.120	0.129	0.165
$\hat{\sigma}(\hat{\beta}_1)/\hat{\sigma}(\hat{\beta}_2)$	0.931	1.242	0.700	0.708
$\hat{\beta}_1$	-0.078	-0.080	-0.037	-0.024
Wald CI	(-0.103, -0.053)	(-0.110, -0.049)	(-0.051, -0.023)	(-0.037, -0.011)
$\hat{\beta}_2$	-0.066	-0.058	-0.043	-0.030
LM CI	(-0.093, -0.039)	(-0.082, -0.033)	(-0.063, -0.024)	(-0.048, -0.012)
$\hat{\beta}^*$	-0.072	-0.064	-0.039	-0.025
Comb. CI	(-0.095, -0.049)	(-0.087, -0.041)	(-0.052, -0.026)	(-0.038, -0.013)

Table 1: Point estimates and confidence intervals: immigrant enclave.

Notes: This table reports the estimation and inference results for the immigrant enclave example using the [Card \(2009\)](#) dataset, shown separately for college equivalent workers and high school equivalent workers. Columns with “Yes” contain city-level controls, while columns with “No” do not. The point estimates are obtained from the standard two-stage least squares (TSLS) estimator with the Bartik instrument, $\hat{\beta}_1$, and, in addition, from the leave-one-out estimator, $\hat{\beta}_2$, which makes use of all base IVs. Wald CI and LM CI denote the confidence intervals based on $\hat{\beta}_1$ and $\hat{\beta}_2$, respectively. The estimator $\hat{\beta}^*$ is the combined estimator for β , defined in Section 4.3. It is essentially the midpoint of the confidence interval in (4.4), which is obtained from our combination test and labeled as “Comb. CI” in the table. In addition, $\hat{\rho}_1$ and $\hat{\rho}_2$ denote estimates of the asymptotic correlation between the Wald and LM statistics, and between the LM and AR statistics, respectively. Finally, $\hat{\sigma}(\hat{\beta}_1)/\hat{\sigma}(\hat{\beta}_2)$ denotes the ratio of standard errors of $\hat{\beta}_1$ and $\hat{\beta}_2$. All displayed numbers are rounded to three decimal places.

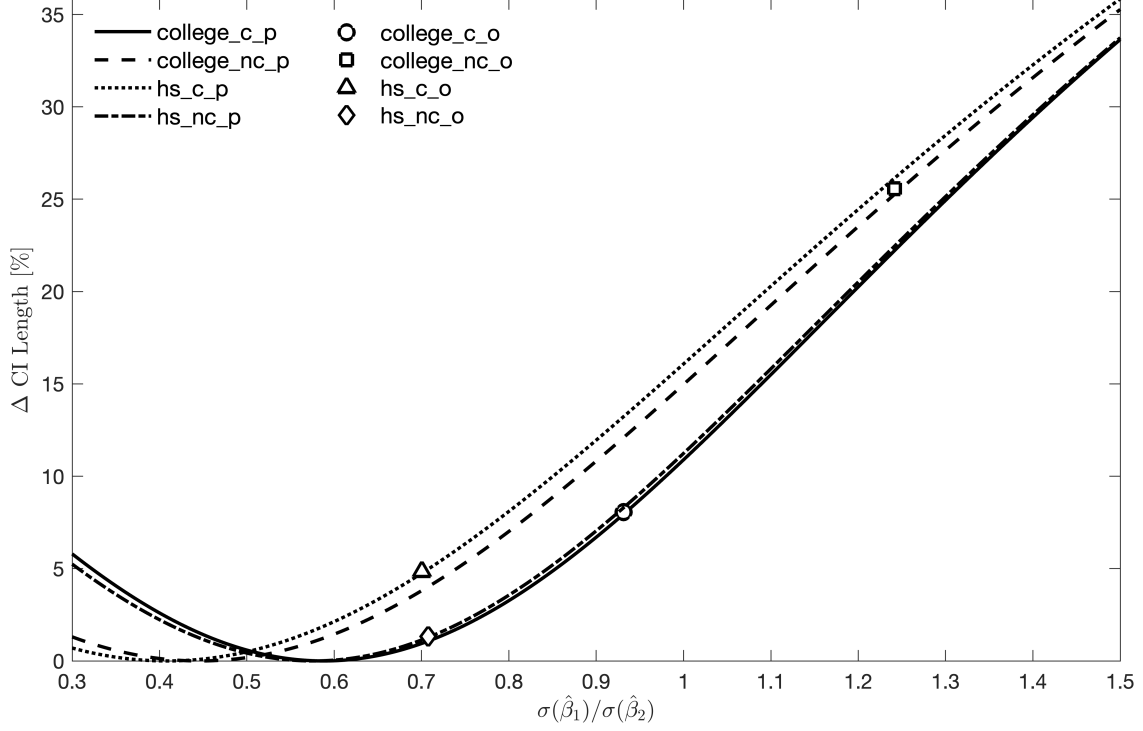


Figure 2: Realized percentage reduction in confidence interval length: immigrant enclave.

Notes: This figure shows, for each specification in the immigrant enclave example, the observed percentage decrease in confidence interval length (Combined CI versus Wald CI, as in Table 1, and indicated by “o” in figure legends) plotted as a point against the standard error ratio ($\hat{\sigma}(\hat{\beta}_1)/\hat{\sigma}(\hat{\beta}_2)$ in Table 1). Also shown is the theoretical lower bound for the reduction (indicated by “p” in figure legends), analogous to Figure 1, but now computed using the specification-specific estimate $\hat{\rho}_1$, as reported in Table 1. Here, “college” refers to the specifications for college equivalent workers, and “hs” refers to the specifications for high school equivalent workers. “c” indicates that controls are included, whereas “nc” indicates that controls are not included. The horizontal axis is the ratio of standard deviations (errors) of $\hat{\beta}_1$ and $\hat{\beta}_2$. The vertical axis is the reduction in the length of confidence interval in percentage points. As a final remark, note that the actual numerical values of the relevant quantities in Table 1, rather than the rounded values shown there, are used to produce Figure 2.

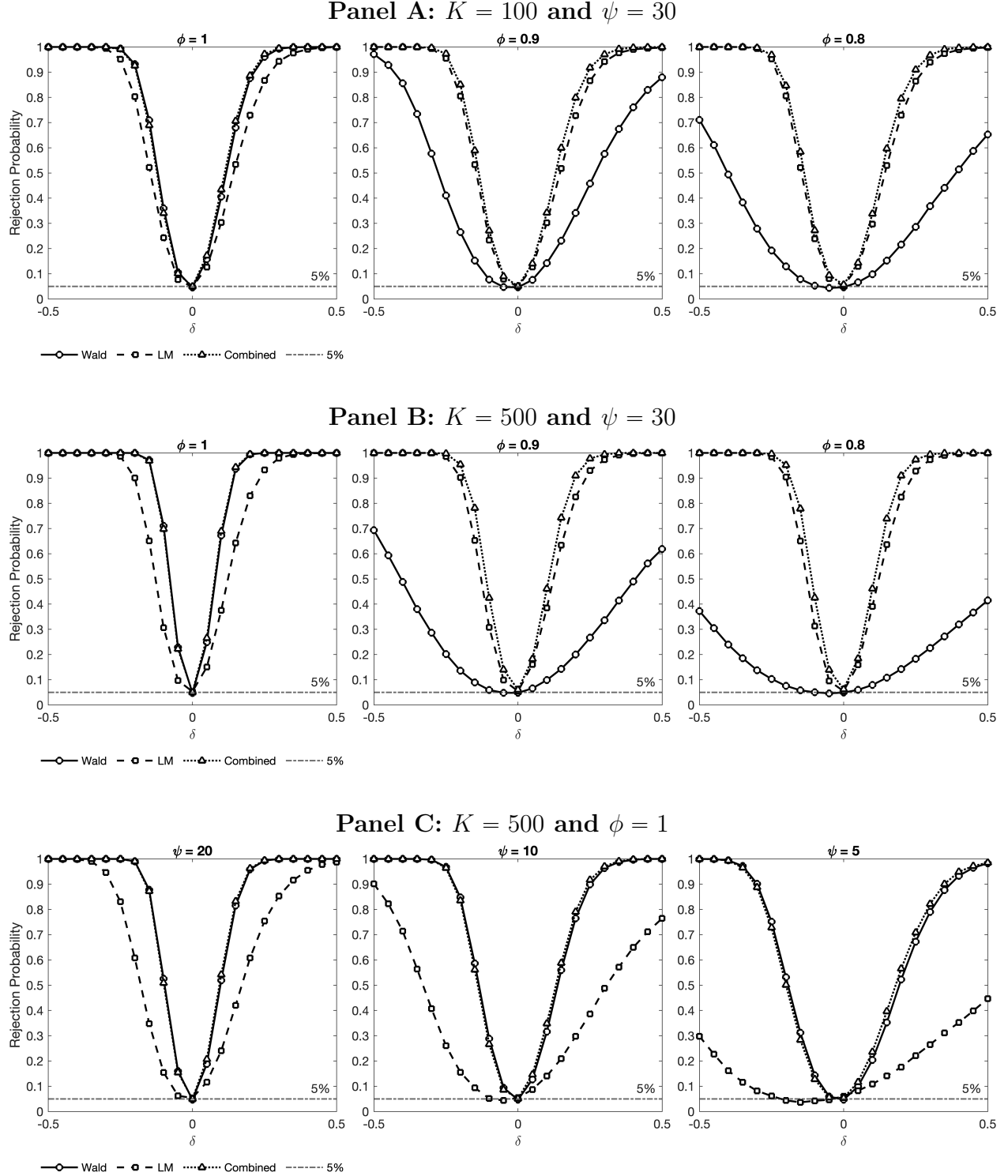


Figure 3: Power curves of the combination, Wald, and jackknife LM tests.

Notes: This figure displays the power curves for our combination test ϕ_n^* along with those for the component Wald and jackknife LM tests, at different values of K (the dimension of the many IVs), ψ (which governs the identification strength of the many IVs), and ϕ (which controls the identification strength of the one-dimensional IV relative to the many IVs). The horizontal axis represents the deviations in the parameter of interest from the maintained hypothesis, that is, we are interested in testing $\mathcal{H}_0 : \beta = \beta_0$ against $\mathcal{H}_1 : \beta \neq \beta_0$, and $\delta = \beta - \beta_0$. See Section 5 for a detailed description of the simulation setup. All results are based on 5,000 simulations.

References

- ACKERBERG, D. A. AND P. J. DEVEREUX (2009): “Improved JIVE estimators for overidentified linear models with and without heteroskedasticity,” *The Review of Economics and Statistics*, 91, 351–362.
- ADAO, R., M. KOLESÁR, AND E. MORALES (2019): “Shift-share designs: Theory and inference,” *The Quarterly Journal of Economics*, 134, 1949–2010.
- ANATOLYEV, S. AND N. GOSPODINOV (2011): “Specification testing in models with many instruments,” *Econometric Theory*, 27, 427–441.
- ANDERSON, T., N. KUNITOMO, AND Y. MATSUSHITA (2010): “On the asymptotic optimality of the LIML estimator with possibly many instruments,” *Journal of Econometrics*, 157, 191–204.
- ANDREWS, D. W. (2017): “Identification-robust subvector inference,” Tech. rep., Cowles Foundation Discussion Paper 2105.
- ANDREWS, D. W. K. AND J. H. STOCK (2007): “Testing with many Weak Instruments,” *Journal of Econometrics*, 138, 24–46.
- ANDREWS, I. (2016): “Conditional linear combination tests for weakly identified models,” *Econometrica*, 84, 2155–2182.
- ANDREWS, I., J. H. STOCK, AND L. SUN (2019): “Weak instruments in instrumental variables regression: Theory and practice,” *Annual Review of Economics*, 11, 727–753.
- ANGRIST, J. D. AND A. B. KRUEGER (1991): “Does Compulsory School Attendance Affect Schooling and Earning?” *Quarterly Journal of Economics*, 106, 979–1014.
- AUTOR, D. H., D. DORN, AND G. H. HANSON (2013): “The China syndrome: Local labor market effects of import competition in the United States,” *American Economic Review*, 103, 2121–2168.

- BARTIK, T. J. (1991): *Who benefits from state and local economic development policies?*, Kalamazoo, MI: WE Upjohn Institute for Employment Research.
- BEKKER, P. (1994): “Alternative approximations to the distributions of instrumental variable estimators,” *Econometrica*, 62, 657–681.
- BELLONI, A., D. CHEN, V. CHERNOZHUKOV, AND C. HANSEN (2012): “Sparse models and methods for optimal instruments with an application to eminent domain,” *Econometrica*, 80, 2369–2429.
- BLANCHARD, O. J., L. F. KATZ, R. E. HALL, AND B. EICHENGREEN (1992): “Regional evolutions,” *Brookings Papers on Economic Activity*, 1992, 1–75.
- BOOT, T. AND D. NIBBERING (2024): “Inference on LATEs with Covariates,” *arXiv preprint arXiv:2402.12607*.
- BORUSYAK, K., P. HULL, AND X. JARAVEL (2022): “Quasi-experimental shift-share research designs,” *The Review of economic studies*, 89, 181–213.
- (2025): “A practical guide to shift-share instruments,” *Journal of Economic Perspectives*, 39, 181–204.
- CARD, D. (2009): “Immigration and Inequality,” *American Economic Review: Papers & Proceedings*, 99, 1–21.
- CARRASCO, M. (2012): “A regularization approach to the many instruments problem,” *Journal of Econometrics*, 170, 383–398.
- CARRASCO, M. AND G. TCHUENTE (2015): “Regularized LIML for many instruments,” *Journal of Econometrics*, 186, 427–442.
- CHAO, J. C. AND N. R. SWANSON (2005): “Consistent Estimation with a Large Number of Weak Instruments,” *Econometrica*, 73, 1673–1692.

- CHAO, J. C., N. R. SWANSON, J. A. HAUSMAN, W. K. NEWEY, AND T. WOUTERSEN (2012): “Asymptotic Distribution Of JIVE In A Heteroskedastic IV Regression With Many Instruments,” *Econometric Theory*, 28, 42–86.
- CHAO, J. C., N. R. SWANSON, AND T. WOUTERSEN (2023): “Jackknife estimation of a cluster-sample IV regression model with many weak instruments,” *Journal of Econometrics*.
- CHYN, E., B. FRANSEN, AND E. C. LESLIE (2024): “Examiner and Judge Designs in Economics: A Practitioner’s Guide,” Tech. rep., National Bureau of Economic Research.
- DAVIDSON, R. AND J. G. MACKINNON (2010): “Wild bootstrap tests for IV regression,” *Journal of Business & Economic Statistics*, 28, 128–144.
- DJOGBENOU, A. A., J. G. MACKINNON, AND M. O. NIELSEN (2019): “Asymptotic theory and wild bootstrap inference with clustered errors,” *Journal of Econometrics*, 212, 393–412.
- DONALD, S. G. AND W. K. NEWEY (2001): “Choosing the number of instruments,” *Econometrica*, 69, 1161–1191.
- DOVÌ, M.-S., A. B. KOCK, AND S. MAVROEIDIS (2024): “A Ridge-Regularized Jackknifed Anderson-Rubin Test,” *Journal of Business & Economic Statistics*, 42, 1083–1094.
- DUFOUR, J.-M. AND M. TAAMOUTI (2005): “Projection-based statistical inference in linear structural models with possibly weak instruments,” *Econometrica*, 73, 1351–1365.
- EVDOKIMOV, K. S. AND M. KOLESÁR (2018): “Inference in Instrumental Variables Analysis with Heterogeneous Treatment Effects,” *Working Paper*.
- FINLAY, K. AND L. M. MAGNUSSON (2019): “Two applications of wild bootstrap methods to improve inference in cluster-IV models,” *Journal of Applied Econometrics*, 34, 911–933.

- FRANDSEN, B., E. LESLIE, AND S. MCINTYRE (2025): “Cluster jackknife instrumental variables estimation,” *The Review of Economics and Statistics*, forthcoming.
- GOLDSMITH-PINKHAM, P., I. SORKIN, AND H. SWIFT (2020): “Bartik instruments: What, when, why, and how,” *American Economic Review*, 110, 2586–2624.
- GUGGENBERGER, P., F. KLEIBERGEN, AND S. MAVROEIDIS (2019): “A more powerful subvector Anderson Rubin test in linear instrumental variables regression,” *Quantitative Economics*, 10, 487–526.
- (2021): “A Powerful Subvector Anderson Rubin Test in Linear Instrumental Variables Regression with Conditional Heteroskedasticity,” *arXiv preprint arXiv:2103.11371*.
- GUGGENBERGER, P., F. KLEIBERGEN, S. MAVROEIDIS, AND L. CHEN (2012): “On the asymptotic sizes of subset Anderson–Rubin and Lagrange multiplier tests in linear instrumental variables regression,” *Econometrica*, 80, 2649–2666.
- HAN, C. AND P. C. PHILLIPS (2006): “GMM with many moment conditions,” *Econometrica*, 74, 147–192.
- HANSEN, C., J. HAUSMAN, AND W. NEWKEY (2008): “Estimation with Many Instrumental Variables,” *Journal of Business and Economic Statistics*, 26, 398–422.
- HANSEN, C. AND D. KOZBUR (2014): “Instrumental variables estimation with many weak instruments using regularized JIVE,” *Journal of Econometrics*, 182, 290–308.
- HAUSMAN, J. A., W. K. NEWKEY, T. WOUTERSEN, J. C. CHAO, AND N. R. SWANSON (2012): “Instrumental variable estimation with heteroskedasticity and many instruments,” *Quantitative Economics*, 3, 211–255.
- JIANG, L., L. LI, K. MIAO, AND Y. ZHANG (2025): “Adjustments with many regressors under covariate-adaptive randomizations,” *Journal of Econometrics*, 249, 105991.

- KOLESÁR, M. (2013): “Estimation in instrumental variables models with heterogeneous treatment effects,” *Working Paper*.
- (2018): “Minimum distance approach to inference with many instruments,” *Journal of Econometrics*, 204, 86–100.
- KUERSTEINER, G. AND R. OKUI (2010): “Constructing optimal instruments by first-stage prediction averaging,” *Econometrica*, 78, 697–718.
- KUNITOMO, N. (1980): “Asymptotic expansions of the distributions of estimators in a linear functional relationship and simultaneous equations,” *Journal of the American Statistical Association*, 75, 693–700.
- LEE, D. S., J. MCCRARY, M. J. MOREIRA, AND J. R. PORTER (2022): “Valid t-ratio Inference for IV,” *American Economic Review*, 112, 3260–3290.
- LEHMANN, E. L. AND J. P. ROMANO (2006): *Testing Statistical Hypotheses*, Springer Science & Business Media.
- LIGTENBERG, J. W. (2023): “Inference in IV models with clustered dependence, many instruments and weak identification,” *arXiv preprint arXiv:2306.08559*.
- LIM, D., W. WANG, AND Y. ZHANG (2024a): “A conditional linear combination test with many weak instruments,” *Journal of Econometrics*, 238, 105602.
- (2024b): “A Dimension-Agnostic Bootstrap Anderson-Rubin Test For Instrumental Variable Regressions,” *arXiv preprint arXiv:2412.01603*.
- MACKINNON, J. G. (2021): “Fast cluster bootstrap methods for linear regression models,” *Econometrics and Statistics*, forthcoming.
- MATSUSHITA, Y. AND T. OTSU (2024): “A jackknife Lagrange multiplier test with many weak instruments,” *Econometric Theory*, 40, 447–470.

- MIKUSHEVA, A. AND L. SUN (2022): “Inference with many weak instruments,” *Review of Economic Studies*, 89, 2663–2686.
- (2024): “Weak identification with many instruments,” *The Econometrics Journal*, 27, C1–C28.
- MORIMUNE, K. (1983): “Approximate distributions of k-class estimators when the degree of overidentifiability is large compared with the sample size,” *Econometrica*, 51, 821–841.
- MÜLLER, U. K. (2011): “Efficient tests under a weak convergence assumption,” *Econometrica*, 79, 395–435.
- NAVJEEVAN, M. (2024): “An Identification and Dimensionality Robust Test for Instrumental Variables Models,” *arXiv preprint arXiv:2311.14892*.
- NEWAY, W. K. AND F. WINDMEIJER (2009): “Generalized method of moments with many weak moment conditions,” *Econometrica*, 77, 687–719.
- OKUI, R. (2011): “Instrumental variable estimation in the presence of many moment conditions,” *Journal of Econometrics*, 165, 70–86.
- SØLVSTEN, M. (2020): “Robust estimation with many instruments,” *Journal of Econometrics*, 214, 495–512.
- STOCK, J. AND M. YOGO (2005): “Asymptotic distributions of instrumental variables statistics with many instruments,” *Identification and inference for econometric models: Essays in honor of Thomas Rothenberg*, 6, 109–120.
- WANG, W. AND F. DOKO TCHATOKA (2018): “On Bootstrap inconsistency and Bonferroni-based size-correction for the subset Anderson–Rubin test under conditional homoskedasticity,” *Journal of Econometrics*, 207, 188–211.
- WANG, W. AND M. KAFFO (2016): “Bootstrap inference for instrumental variable models with many weak instruments,” *Journal of Econometrics*, 192, 231–268.

WANG, W. AND Y. ZHANG (2024): “Wild bootstrap inference for instrumental variables regressions with weak and few clusters,” *Journal of Econometrics*, 241, 105727.

YAP, L. (2024): “Inference with Many Weak Instruments and Heterogeneity,” *arXiv preprint arXiv:2408.11193*.